

# SCI-FLT-FIXED v0.1 Shared Normative Scientific Core

SCI-FLT-FIXED-NORMATIVE-CORE v0.1/draft-r0.2

**Status: implementation-blind Stage B scientific-contract draft; scientific-owner review required**

**Scientific owner: Grant Wilson**

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```

Implementation-blind scientific-contract draft. This artifact reports no implementation, validation, calibration, achieved-response, achieved-covariance, performance, readiness, freeze, production, or Unity result.

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## 1. Authority, scope, and normative language

This document is the sole shared normative scientific core for the SCI-FLT-FIXED v0.1 Stage B draft. Revision r0.2 incorporates the exact scientific-owner formal-closure directive bound by the Stage B build record. It preserves the scientific authority of the exact 17-object packet bound by AUTHOR\_PACKET\_MANIFEST.md identity SCI-FLT-FIXED\_AUTHOR\_PACKET v0.1/r0.1. The external SHA-256 of that manifest is 7f2d03f182258ac9770f7dba869e9ae0b5018efdcdb93b18b299a9b9c6df1e4d.

The key words MUST, MUST NOT, REQUIRED, SHALL, SHALL NOT, SHOULD, SHOULD NOT, MAY, and OPTIONAL state the strength of a scientific-contract requirement. Unknown, unavailable, disabled, failed, and zero are distinct typed states.

SCI-FLT is the recovery tranche. SCI-FLT-FIXED is the v0.1 package. No SCI-FLT-INF contract is defined here.

## 2. Scientific object and estimand

For one exact admitted parent map vector  $m$ , the scientific object is

```
J_full = J(plan,
            immutable parent domain,
            FLT input admission,
            availability,
            finiteness,
            support,
            required predicates).

A_Theta,J = J_full L_Theta,
y = A_Theta,J m.
```

The only numerically admitted base family is FLT-FIXED-CONV.  $L_{\text{Theta}}$  is the complete matrix representation of one exact realized sampled convolution; it is not a separately selectable arbitrary dense linear-operator family. FLT-FIXED-CONV-LOWPASS is only a qualified subtype of FLT-FIXED-CONV.

$L_{\text{Theta}}$  and the externally resolved plan are complete and frozen before application to the parent random field.  $J_{\text{full}}$  is parent-state-dependent but is resolved once from immutable parent membership and the listed predicates before payload arithmetic. The resulting  $A_{\text{Theta},J}$  is the complete applied operator. The output  $y$  is transformed parent-map amplitude on the selected row domain. There is no additive term.

Strict linearity is conditional on the exact frozen parent membership and  $J_{\text{full}}$ . It is not a global-linearity claim across parents whose domain, support, admission, availability, finiteness, or validity differs. Response, covariance, noise realizations, and NOI members receive the identical frozen  $A_{\text{Theta},J}$ ;  $J_{\text{full}}$  is never re-resolved for a response perturbation, covariance draw, noise realization, or fixed-state NOI member. A member that cannot supply one frozen required footprint is unavailable on the affected row rather than receiving another selector. Selection and support uncertainty are excluded from conditional linear propagation unless separately supplied as a typed uncertainty object.

Fixed convolution is the admitted structured family. Fixed low-pass convolution is only a qualified subtype of fixed convolution. It is not a second generic family and it is not implied by a friendly name, a width, or the word smoothing.

## 3. Parent roles and parent ordering

One transformation binds exactly one immutable parent in exactly one role:

- FLT-PARENT-MAP-OBS: one complete base or unfiltered MAP observation bundle;
- FLT-PARENT-MAP-COADD: one complete base or unfiltered centered-integer common-grid MAP coadd bundle; or
- FLT-PARENT-JINC-OBS: one complete atomic JINC observation bundle.

The roles are not interchangeable even when shape, WCS values, or numerical payloads happen to agree. MAP observation, MAP coadd, and JINC observation successors remain separately identified. SCI-FLT-FIXED does not coadd and does not presume that filtering and coaddition commute.

The exact parent binding carries the complete parent package, revision, product and application generations, role, membership, stable array or group, quantity, units and originating nominal-beam convention, WCS, frame, topology, grid, pixel metric, shape, row domain, support, validity and causes, response and covariance availability, null and additive-reference state, exposure, causal influence, lifecycle, failure, and provenance facts required by the applicable MAP or JINC boundary. For JINC, all five required numerical roles form one atomic parent. A partial bundle is not a parent.

The Stage A boundary records make ordinary numerical MAP and ToI TEC JINC parents unavailable at this draft's launch state. The contract defines their typed successor routes but does not manufacture numerical parents.

## 4. Requested, effective, resolved, and applied state

The contract distinguishes:

- requested scientific purpose and method;
- effective selection or disablement;
- complete externally resolved immutable plan;
- observation-resolved operator generation;
- exact frozen row selector and applied transformation;
- complete publication candidate;
- publication decision; and
- realized atomic successor product.

Every coefficient, kernel parameter, cutoff or width, WCS fact, normalization, support rule, transfer qualification, and lifecycle fact is resolved before application under one exact FLT-owned plan and fixed before use. "Externally resolved" describes timing and separation from payload arithmetic; it does not transfer FLT policy or transformation ownership to an unnamed producer. No parent data, response perturbation, covariance draw, noise realization, or NOI member may select, learn, tune, or re-resolve any plan or selector fact.

## 5. Same-grid finite operator

Let  $S_{in}$  be the exact finite parent row domain. For output row  $p$ ,

$$y_p = \text{sum over } q \text{ in } S_{in} \text{ of } L[p,q] m_q, \text{ for } p \text{ in } S_{out}.$$

One SCI-FLT-FIXED product applies one exact resolved sampled convolution. A plan may supply a final kernel constructed elsewhere, but the FLT product makes no claim about intermediate transformations. Ordered composition, multiple convolution application, and implicit operator reordering are not admitted in base v0.1. In particular, no equality between a sequence with intermediate selectors and one final selected convolution is presumed.

The operator identity binds all of the following:

- exact input and output row domains;
- identical parent and output WCS, frame, topology, metric, shape, indexing, and pixel-area convention, apart from restriction to  $S_{out}$ ;
- operator family, version, parameter set, complete sampled coefficients, and content digest;
- coefficient coordinate domain, units, ordering, numerical representation, orientation, handedness, center, extent, even or odd tie convention, phase, subpixel convention, coordinate-domain method, and finite support sets;
- normalization and every qualified transfer fact;
- full-footprint row selection and unavailable-row cause rules;
- response, covariance, mode, influence, support, and validity states; and
- requested through realized lifecycle generation, failure, and provenance.

Reprojection, resampling, mosaicking, and deconvolution are not same-grid SCI-FLT-FIXED methods. FFT, direct, separable, cached, or threaded evaluation is scientifically immaterial only when it realizes the identical declared finite operator under a preregistered numerical-comparison policy.

## 6. Fixed convolution and low-pass qualification

For the exact finite offset representation, fixed convolution is

$$(L\_Theta\ m)_p = \text{sum over } r \text{ in } K\_geom \text{ of } k\_Theta(r)\ m_{(p-r)}.$$

The sampled kernel, rather than a continuous ideal or family name, constructs the scientific operator. Every admitted kernel declares three exact sets:

- $K\_geom$ : the declared geometric or storage footprint;
- $K\_nonzero$ : offsets whose declared sampled coefficient is exactly nonzero; exact zero is established by the coefficient representation, never by a floating threshold; and
- $K\_req$ : parent locations scientifically required by the method.

For an ordinary admitted fixed convolution,  $K\_nonzero$  is a subset of  $K\_req$ , and  $K\_req$  is a subset of  $K\_geom$ ; the plan states whether any exact geometric zero is nevertheless required. Base v0.1 chooses the narrow rule that every stored geometric coefficient location is scientifically required, so  $K\_req = K\_geom$ , including exact-zero stored coefficients. A sparse coefficient representation may omit geometric zeros only when its declared  $K\_geom$  and  $K\_req$  preserve the same method identity. Geometric, nonzero, required-dependency, signed, absolute, and squared support remain distinct objects.

The identity convolution has  $K\_req = \{0\}$ . Its scientific row domain is the exact admitted finite parent-row domain. The exact zero convolution does not receive all storage rows through a vacuous empty-footprint predicate: it binds and inherits the exact admitted finite parent-support row domain as  $K\_req$  domain authority while producing numerical zero there.

A convolution declares one exact coordinate-domain method: pixel-index domain, affine tangent-plane angular domain, or another exact identified domain. A fixed shift-invariant pixel kernel is not one spatially uniform angular kernel unless that relation is established over the declared WCS.

An angular-frequency low-pass claim is available only when the resolved plan binds either one exact affine or constant tangent-plane pixel metric, or another exact method proving the stated angular transfer. It also binds the exact frequency grid and domain, DC gain, passband, transition region, stopband or attenuation criterion, phase, isotropic or anisotropic state, finite-grid and edge limitations, sampled kernel, normalization, and parameter identity, source, and provenance. If any one is unavailable, fixed-convolution identity may remain available but low-pass qualification is unavailable.

The sampled-kernel or interior translation-invariant transfer is distinct from the complete finite row-restricted operator  $A_{\text{Theta}, J}$ . Base v0.1 makes no global Fourier-transfer claim for that complete finite operator unless an exact theorem bound to its finite domain and selector establishes one.

## 7. Full-footprint scientific output domain

The sole v0.1 scientific output-row method is

```
S_out = {p: for every r in K_req,
            p-r lies in S_in,
            m_(p-r) is admitted for this exact FLT use,
            m_(p-r) is available and finite,
            and every required predicate passes}.
```

$K_{\text{req}}$ , not merely nonzero coefficient support or storage shape, governs scientific admission. Rows outside  $S_{\text{out}}$  are scientifically unavailable, not zero. A stored array may preserve parent shape and WCS only if every unavailable row carries its typed cause and remains outside the scientific vector.

Base v0.1 admits no boundary extension, periodic wrapping, truncated convolution, support-conditioned renormalization, inpainting, reflection, clamping, mirroring, edge completion, padding-based admission, or replacement of a missing or non-finite value. Each would require a separately named and versioned successor method.

Numerical computability, complete kernel support, FLT input admission, FLT-local output validity, confidence, and downstream eligibility are distinct states.

## 8. Quantity, units, beam, response, and modes

FLT-SIG is the transformed parent-map quantity. Its units are derived from the exact parent units and operator coefficient units. For a MAP parent, the originating mJy/beam nominal-beam and calibration lineage remain identified; the output is not relabeled as mJy/filtered-beam.

Kernel normalization is an explicit convention, such as unit signed sum or DC gain, unit peak, unit angular integral, unit L2 norm, or another fully stated convention. No convention is inferred from a name. Numerical unit preservation does not establish calibration, point-source peak preservation, integrated flux, extended-source fidelity, target PSF, or a new beam convention.

Response families are separately typed and none may silently stand for another:

- Fixed-state linear parent response:

```
R_out^fixed = A_Theta, J R_parent^fixed;
```

- An already realized parent-grid response companion receives  $A_{\text{Theta}, J}$  exactly once.
- A parent full-procedure finite difference with FLT fixed follows:

```
Delta y_parent-FP = A_Theta, J Delta m_parent-FP,
```

The parent state-change record is retained separately.

- An FLT re-resolved procedure response is outside SCI-FLT-FIXED when kernel, cutoff, normalization, support, selector, edge state, or method is re-resolved.

Every admitted response uses the identical frozen  $A_{\text{Theta}, J}$ , centering, orientation, phase, support, edge, missing-data, row-domain, and normalization rules as the signal. If its required source basis, domain, or compatible parent-response identity is unavailable, the requested FLT-RSP family is typed unavailable. The kernel alone is not the complete source response or PSF.

For the exact zero operator, the local fixed-state derivative and the parent-payload conditional covariance contribution are exactly zero on its declared scientific row domain. A complete source-domain response may still be unavailable when its source basis, domain, or parent-response identity is unavailable. Total

systematic uncertainty remains separately typed. Local zero response is neither hidden nor promoted into an unsupported complete sky-response claim.

FLT-TRANSFER records the exact local sampled spatial or Fourier transfer on the declared finite-grid domain where scientifically defined, or typed unavailability. FLT-MODE records exact null modes, invariant modes, attenuation statements or bounds, and sampled phase where defined. Local operator modes are not automatically upstream sky-to-output modes.

FLT-INFLUENCE is the exact coefficient and support relation between parent and output rows. It is not physical exposure. Filtering creates no new physical exposure claim. FLT-EXPOSURE-LINEAGE records the exact parent exposure-product identity or typed absence and states that FLT creates no physical exposure. Convolving an exposure plane is not authorized as the physical exposure of filtered signal.

## 9. Deterministic covariance and NOI boundary

For an available compatible declared parent covariance  $C_{\text{parent}}$ ,

$$C_{\text{out}} = A_{\text{Theta},J} C_{\text{parent}} \text{ transpose}(A_{\text{Theta},J}).$$

Parent stochastic authority and output representation are separate axes. Parent authority is exactly one of complete covariance, explicit independent-diagonal model, marginal variances only, structured or partial model, or unavailable. Output representation is exactly one of complete matrix, exact linear or operator representation, structured representation, marginal plane, summary only, or unavailable.

Every result states the parent model on which it is conditional, output representation, domain and ordering, rank and null space, omitted terms, supported operations, and selection, kernel, beam, WCS, calibration, and model uncertainty exclusions. A full matrix propagated from an explicitly declared independent-diagonal parent model is complete relative to that conditional model; it does not imply that unknown real parent cross terms are zero.

For an explicitly diagonal independent parent covariance with marginal variances  $V_j$ ,

$$\begin{aligned} A_{ij} &= (J_{\text{full}} L_{\text{Theta}})_{ij}, \\ \text{Var}(y_i) &= \sum_j A_{ij}^2 V_j, \\ \text{Cov}(y_i, y_l) &= \sum_j A_{ij} A_{lj} V_j. \end{aligned}$$

The output generally has off-diagonal covariance. A marginal variance plane is not full covariance and does not authorize independent-pixel multi-pixel inference.

SCI-NOI, not SCI-FLT-FIXED, owns empirical uncertainty, empirical covariance, conditional inverse scale, standardized signal, and significance inference. For every compatible admitted NOI member  $M_b$ , exact fixed-state parity is

$$M_{b,\text{out}} = A_{\text{Theta},J} M_b.$$

The real parent and every member use the identical frozen  $A_{\text{Theta},J}$  parameter set, grid, support, edge rule, row domain, and lifecycle generation.  $J_{\text{full}}$  is never re-resolved per member. Filtering a variance, standard deviation, precision, reciprocal, weight, standardized map, or significance field is not this operation. Per-member selection or re-resolution is a different inference-bearing method and cannot enter the fixed-state ensemble.

Parameter, kernel, cutoff, beam, WCS, selection, model, and calibration uncertainty remain separate from covariance conditional on fixed  $A_{\text{Theta},J}$  unless a separately supplied typed model includes them.

## 10. Atomic product and lifecycle

Every realized bundle contains these role records, including explicit unavailable states where allowed:

- FLT-PARENT;
- FLT-PLAN;

- FLT-OPERATOR;
- FLT-SIG;
- FLT-UNIT-BEAM;
- FLT-TRANSFER;
- FLT-RSP;
- FLT-MODE;
- FLT-INFLUENCE;
- FLT-EXPOSURE-LINEAGE;
- FLT-SUP;
- FLT-VALID;
- FLT-COV-FORMAL;
- FLT-NOI-ATTACHMENT-STATE; and
- FLT-LINEAGE.

A missing required record is not an atomic bundle. An honestly unavailable response or covariance record may satisfy the base role only where the role permits absence; it cannot satisfy a response-qualified or covariance-qualified request.

FLT-NOI-ATTACHMENT-STATE is a relation record, never the NOI product. It contains the exact FLT product identity, compatible fixed-operator and row-domain identity, whether attachment was requested, an attached NOI product identity only if separately realized later, and otherwise a typed unavailable, not\_requested, or failed state. The separately parented and separately realized SCI-NOI product is not required for FLT atomic completion and never changes whether the FLT bundle is complete or realized.

The publication lifecycle is

```

requested
-> effective
-> resolved
-> applied
-> complete_publication_candidate
-> publication_decision
-> realized | failed | not_produced.
```

SCI-FLT-FIXED:output\_publication@1 evaluates the complete publication candidate, not a pre-existing realized bundle. A successful publication action creates realized\_identity, realized\_zero, or realized as applicable. Disabled is not\_produced. Failure of a required transformation, candidate, or publication step propagates and creates no complete product. unavailable remains a typed route or role state rather than a synonym for disabled, failed, zero, or not produced. superseded records immutable succession after a realized generation exists.

Any change to parent, request or effective purpose, operator, kernel, parameter, transfer qualification, normalization, WCS, grid, row domain, support, validity, response or covariance role, lifecycle, or failure policy creates a new immutable transformation and product generation. A later NOI attachment is a separate immutable companion and does not mutate FLT or the parent.

## 11. Typed FLT policy objects and VAL boundary

The draft defines three distinct policy domains:

- SCI-FLT-FIXED:input\_bundle\_admission@1 evaluates one request, one complete parent bundle, and one exact resolved FLT plan;

- SCI-FLT-FIXED:input\_row\_admission@1 evaluates one candidate output row against the frozen plan, parent domain, K\_req, availability, finiteness, support, and required predicates and contributes to the single J\_full; and
- SCI-FLT-FIXED:output\_publication@1 evaluates one complete publication candidate and performs or denies the publication action.

Every immutable profile binds scientific-policy owner, exact source and boundary versions, request, applicability, eligibility, realization, restrictions, decisive exclusions, exceptions, missing or conflict behavior, lifecycle, and exact consumer action. The exact draft records are part of the bound Stage B source set.

Structural identity or source conflict yields applicability\_unknown and decision\_unavailable. A known applicable failed restriction yields ineligible. A not-requested or effectively disabled route makes no eligibility proposition and yields not\_produced. An eligible transformation failure yields eligible and realization\_failed. An eligible complete publication candidate becomes realized only after successful publication action.

Companion qualification is request-specific:

- a base signal request requires exact response and covariance state records but permits an honest unavailable state where their role allows it;
- a response-qualified request requires an available exact compatible response;
- a covariance-qualified request requires the exact declared compatible covariance representation; and
- a response-and-covariance-qualified request requires both.

An optional unavailable companion never blocks an otherwise valid base-signal product. It does block the corresponding qualified request.

SCI-VAL may bind and evaluate an immutable owner-approved successor of these profiles. VAL does not author producer facts, FLT policy, arithmetic, or scientific claims. The r0.2 profile bytes and source bindings are drafts, are not owner-approved Registry entries, and create no claim that Registry evaluation occurred.

## 12. Consumer and ownership boundaries

MAP and JINC own the parent estimand and parent claims. CAL owns absolute calibration, passband and color correction, and calibration covariance. SCI-FLT-FIXED owns the exact local transformation, transformed signal, output unit derivation, composed-response state, local transfer and modes, influence, support and validity, deterministic covariance state, lifecycle, failure, and provenance. It also owns its plan, selector, application, and publication policy even when a final kernel is constructed elsewhere. SCI-NOI owns empirical uncertainty inference and applies but does not choose the exact FLT transformation. SCI-BEAM and future source or mode contracts own physical source, beam, Pointing, and OOF interpretations. SCI-FRUIT owns iterative feedback science.

confidence is not a generic FLT scalar. Where the applicable upstream boundary supplies a named confidence or quality state, FLT carries its exact typed identity and meaning as parent state and may reference it in a named row predicate. Otherwise FLT confidence is not\_defined; it must not be inferred from finiteness, support, covariance, a weight, or downstream eligibility.

The availability of an SCI-FLT-FIXED product authorizes no generic Beammap, Pointing, OOF, source-fit, catalog, NOI, or FRUIT use. Each consumer owns an exact use policy.



## 13. Numerical-conformance policy boundary

Future finite-precision implementation evidence must be evaluated under one preregistered immutable numerical-comparison policy. Before observing a candidate result, that policy binds an independent exact or high-precision oracle; absolute, relative, and near-zero behavior; signed-cancellation and zero-sum cases; conditioning and operation-count dependence; covariance comparison; sequential and parallel agreement; overflow, underflow, and non-finite handling; simultaneous row-level comparison; lifecycle; and provenance. Bounds must not be changed after observing a failure.

The policy preserves exact operator coefficients and scientific identity. It authors no filter science and supplies no implementation-conformity, validation, numerical-adequacy, or performance claim. The r0.2 draft policy is bound as a separate Stage B source artifact and is not a preregistered future evidence decision.

## 14. Exclusions and nonclaims

This contract excludes affine offsets, background or template subtraction, additive correction, reprojection, resampling, mosaicking, deconvolution, boundary extension, truncated or renormalized convolution, missing-value replacement, data-derived kernel or cutoff selection, automatic method selection, Wiener transformation, matched or generalized least-squares template-amplitude estimation, source-learned operation, data-derived mode selection or map-domain destriping, per-member relearning, FLT coaddition, source or mode interpretation, FRUIT recurrence, and RTC timestream filtering.

This Stage B draft makes no implementation-conformity, algorithm-change, validation, calibration, achieved-response, achieved-covariance, numerical adequacy, performance, readiness, scientific-freeze, production, or Unity claim.

## 15. Normative requirements

### SCI-FLT-FIXED-REQ-001 - Package identity

The package SHALL be identified as SCI-FLT-FIXED v0.1 within the SCI-FLT tranche. FLT-FIXED-CONV SHALL be the only numerically admitted base family, with FLT-FIXED-CONV-LOWPASS only a qualified subtype. It SHALL NOT be presented as SCI-FLT-INF, an arbitrary dense linear-operator family, or a generic filter contract.

### SCI-FLT-FIXED-REQ-002 - Strict linearity

The transformation MUST be exactly  $y = A_{\text{Theta}, J} m$ , where  $A_{\text{Theta}, J} = J_{\text{full}} L_{\text{Theta}}$ , with no additive term. Strict linearity MUST be conditioned on one exact frozen parent membership and selector. Any offset, background, template subtraction, or additive correction MUST be rejected as outside v0.1.

### SCI-FLT-FIXED-REQ-003 - Exact parent role

Each transformation MUST bind exactly one complete immutable FLT-PARENT-MAP-OBS, FLT-PARENT-MAP-COADD, or FLT-PARENT-JINC-OBS parent. No role substitution by shape, WCS, name, or payload similarity is permitted.

### SCI-FLT-FIXED-REQ-004 - Complete parent identity

The parent binding MUST include all applicable package, revision, product, application, observation or membership, quantity, unit, nominal-beam, WCS, grid, row-domain, support, validity, response, covariance, null, exposure, lifecycle, failure, and provenance facts stated in Section 3. A partial JINC five-role bundle MUST be rejected.

### SCI-FLT-FIXED-REQ-005 - Honest upstream availability

An unavailable MAP or JINC numerical parent MUST remain unavailable. Algebra, defaults, approximate identity, or a finite payload MUST NOT manufacture a parent or successor.

### **SCI-FLT-FIXED-REQ-006 - Parent ordering**

Filtering an observation and filtering a coadd MUST create distinct successor identities. SCI-FLT-FIXED MUST NOT coadd or claim filter/coadd commutation without a separately approved exact bounded relation.

### **SCI-FLT-FIXED-REQ-007 - External fixed resolution**

Every operator coefficient, parameter, grid fact, normalization, support rule, and transfer qualification MUST be resolved under one exact FLT-owned plan and frozen before application. The selector MUST be resolved once before payload arithmetic and MUST NOT be re-resolved for response, covariance, noise, or NOI members.

### **SCI-FLT-FIXED-REQ-008 - Same-grid boundary**

The output MUST preserve the exact parent WCS, frame, topology, metric, shape, pixel indexing, and pixel-area convention, apart from scientific-row restriction. Reprojection, resampling, and approximate-WCS joins MUST be rejected.

### **SCI-FLT-FIXED-REQ-009 - Complete operator identity**

The applied operator MUST bind every identity and discretization fact listed in Sections 5 and 6, including complete sampled coefficients, K\_geom, K\_nonzero, K\_req, coordinate-domain method, and content digest. A friendly kernel name or continuous ideal is insufficient.

### **SCI-FLT-FIXED-REQ-010 - Fixed convolution construction**

A FLT-FIXED-CONV method MUST construct the complete finite operator from one exact finite sampled kernel and its declared offset sets, orientation, center, phase, normalization, coordinate-domain method, and coefficient representation. One product MUST apply this convolution exactly once and MUST make no intermediate transformation or reordered-composition claim.

### **SCI-FLT-FIXED-REQ-011 - Low-pass qualification**

A FLT-FIXED-CONV-LOWPASS claim MUST bind every coordinate, metric, frequency, and transfer fact listed in Section 6 and distinguish sampled-kernel transfer from the complete finite row-restricted operator. If any fact is missing, the low-pass claim MUST be unavailable even when fixed-convolution identity remains available.

### **SCI-FLT-FIXED-REQ-012 - Full-footprint admission**

S\_out MUST use K\_req and contain exactly those rows for which every scientifically required parent location is in the parent domain, admitted for the exact FLT use, available, finite, and passing every required predicate. Exact-zero coefficient status MUST be read from the declared representation, never a floating threshold.

### **SCI-FLT-FIXED-REQ-013 - Unavailable row semantics**

A row outside S\_out MUST be unavailable rather than zero. Parent-shaped storage MAY be retained only with explicit unavailable-row state and causes.

### **SCI-FLT-FIXED-REQ-014 - Sole edge method**

Boundary extension, wrapping, truncation, support renormalization, inpainting, reflection, clamping, mirroring, padding-based admission, edge completion, and missing or non-finite replacement MUST NOT be used by v0.1.

### **SCI-FLT-FIXED-REQ-015 - Output quantity and units**

FLT-SIG MUST be identified as transformed parent-map amplitude on S\_out, with output units derived from exact parent and coefficient units. It MUST NOT be relabeled as flux, fitted amplitude, or a response-corrected quantity.

### SCI-FLT-FIXED-REQ-016 - Beam and calibration boundary

The originating nominal-beam and calibration lineage MUST be retained. No new filtered-beam, target-PSF, absolute-calibration, peak-preservation, integrated-flux, or extended-source claim may be inferred from normalization or units.

### SCI-FLT-FIXED-REQ-017 - Response composition

Each FLT-RSP MUST identify exactly one response family in Section 8. An available compatible fixed-state response, realized parent-grid response, or parent full-procedure finite difference MUST receive the identical frozen  $A_{\text{Theta}, J}$  exactly once as specified. FLT re-resolution is outside v0.1. A missing required basis, domain, or parent-response identity MUST yield typed unavailability rather than substitution.

### SCI-FLT-FIXED-REQ-018 - Transfer and mode state

FLT-TRANSFER and FLT-MODE MUST publish the exact sampled-kernel or local finite-grid transfer, null, invariant, attenuation, and phase facts where defined, or honest typed unavailability. They MUST NOT be promoted to an unproved global finite-operator, angular-WCS, or whole-chain claim.

### SCI-FLT-FIXED-REQ-019 - Influence is not exposure

FLT-INFLUENCE MUST describe the exact parent-to-output coefficient relation and MUST NOT be labeled or interpreted as physical exposure. FLT-EXPOSURE-LINEAGE MUST identify the exact parent exposure product or typed absence and MUST state that FLT creates no physical exposure.

### SCI-FLT-FIXED-REQ-020 - Distinct support and validity states

Numerical computability,  $K_{\text{geom}}$ ,  $K_{\text{nonzero}}$ ,  $K_{\text{req}}$ , complete footprint, FLT bundle and row admission, FLT-local validity, parent validity, named upstream confidence or  $\text{not\_defined}$ , and downstream eligibility MUST remain distinct with exact causes.

### SCI-FLT-FIXED-REQ-021 - Deterministic covariance propagation

An available compatible parent covariance MUST be propagated by  $C_{\text{out}} = A_{\text{Theta}, J} C_{\text{parent}} \text{transpose}(A_{\text{Theta}, J})$  on the exact scientific row domain with the identical frozen selector.

### SCI-FLT-FIXED-REQ-022 - Covariance state honesty

FLT-COV-FORMAL MUST separately state parent stochastic authority and output representation using Section 9's two axes, plus conditional model, domain, ordering, rank, null space, omissions, supported operations, and excluded uncertainties. Unknown cross terms MUST NOT be set to zero or interpreted as independence.

### SCI-FLT-FIXED-REQ-023 - Induced covariance

For an explicitly diagonal independent parent model, the exact output off-diagonal terms using  $A_{ij}$  MUST be retained in any complete-relative-to- that-model covariance claim. A marginal plane MUST NOT be called full covariance, and the declared diagonal model MUST NOT imply unknown real parent cross terms are zero.

### SCI-FLT-FIXED-REQ-024 - Empirical uncertainty ownership

SCI-FLT-FIXED MUST NOT infer empirical uncertainty, empirical covariance, conditional inverse scale, standardized signal, or significance. Such an attachment remains SCI-NOI-owned and separate from FLT-COV-FORMAL.

### SCI-FLT-FIXED-REQ-025 - Fixed-state NOI parity

Every admitted NOI member for the exact transformed product MUST receive the identical frozen  $A_{\text{Theta}, J}$ , parameters, grid, support, edge rule, row domain, and lifecycle generation used for the real parent. A member failing a frozen required footprint MUST be unavailable there rather than receiving a new selector.

### **SCI-FLT-FIXED-REQ-026 - Relearning rejection**

Any per-member parameter selection, operator resolution, or relearning MUST be rejected from the fixed-state ensemble and routed to a separately named inference-bearing method.

### **SCI-FLT-FIXED-REQ-027 - Atomic product roles**

A realized product MUST contain every required role record in Section 10, including explicit unavailable companion records where the base role permits absence. FLT-NOI-ATTACHMENT-STATE, not an SCI-NOI product, MUST occupy the FLT relation role. A partial bundle MUST NOT be published.

### **SCI-FLT-FIXED-REQ-028 - Lifecycle**

The lifecycle MUST evaluate publication policy against a complete publication candidate before realization and MUST distinguish all states and transitions in Section 10 while preserving exact causes, failures, and immutable generation bindings.

### **SCI-FLT-FIXED-REQ-029 - Disabled, identity, and zero states**

Disabled MUST be not\_produced. Requested and applied identity and zero operators MUST use their explicit support rules, complete candidate, and publication decision to produce distinct realized products and MUST NOT be represented as disabled or unavailable. Zero MUST NOT acquire rows through a vacuous empty-footprint test.

### **SCI-FLT-FIXED-REQ-030 - Generation identity**

Any change listed in Section 10 MUST create a new immutable transformation and product generation. A separately realized NOI product or later attachment relation MUST NOT determine FLT completeness or mutate the FLT product.

### **SCI-FLT-FIXED-REQ-031 - Failure and fallback**

Missing, conflicting, or unavailable state required by the requested identity MUST fail the affected route closed with exact causes. No silent fallback, default, or same-name substitution may retain that requested identity.

### **SCI-FLT-FIXED-REQ-032 - Input admission policy**

The SCI-FLT-FIXED:input\_bundle\_admission@1 and SCI-FLT-FIXED:input\_row\_admission@1 draft semantics in Section 11 MUST govern their distinct bundle and row domains, typed applicability and eligibility, unavailable decision, exclusions, and fail-closed action until superseded by owner-approved immutable profiles.

### **SCI-FLT-FIXED-REQ-033 - Output publication policy**

The SCI-FLT-FIXED:output\_publication@1 draft semantics in Section 11 MUST govern the exact complete publication candidate and successful publication action. Honest absence is permitted only for a role and request qualification that explicitly allows it.

### **SCI-FLT-FIXED-REQ-034 - VAL boundary**

SCI-VAL MAY bind and evaluate an owner-approved immutable policy but MUST NOT author FLT facts or policy, perform the transformation, or convert these unapproved draft profile bytes into a Registry-evaluated or numerical route.

### **SCI-FLT-FIXED-REQ-035 - Consumer boundary**

Product availability MUST NOT authorize generic downstream use. Beammap, Pointing, OOF, source-fit, catalog, NOI, and FRUIT consumers MUST supply their own exact owner-approved use policies.

**SCI-FLT-FIXED-REQ-036 - Excluded methods and nonclaims**

Every method in Section 14 MUST remain outside v0.1. The package MUST preserve all Stage B nonclaims in Section 14.

**SCI-FLT-FIXED-REQ-037 - Frozen-selector conditioning**

$J_{full}$  MUST be resolved exactly once before payload arithmetic from the immutable parent membership and required predicates. Response perturbations, covariance draws, noise realizations, and NOI members MUST reuse the resulting  $A_{\Theta, J}$ ; selection and support uncertainty MUST remain excluded unless separately supplied as typed uncertainty.

**SCI-FLT-FIXED-REQ-038 - Required-dependency support**

Every kernel MUST distinguish  $K_{geom}$ ,  $K_{nonzero}$ , and  $K_{req}$  and bind their exact relations. Base-v0.1 stored geometric zeros remain required; identity uses  $K_{req} = \{0\}$ ; and zero inherits a declared admitted parent-support row domain rather than a vacuous empty-footprint domain.

**SCI-FLT-FIXED-REQ-039 - Typed policy and companion qualification**

All three policy objects in Section 11 MUST bind their exact domains and VAL- congruent state semantics. Base, response-qualified, covariance-qualified, and jointly qualified requests MUST apply their distinct companion requirements; an optional unavailable companion MUST NOT block a valid base signal.

**SCI-FLT-FIXED-REQ-040 - Response-family separation**

Fixed-state linear response, already realized parent-grid response, parent full-procedure finite difference with FLT fixed, and FLT re-resolved procedure response MUST remain separately identified. No response family or local zero derivative may be promoted into another or into an unsupported complete source-domain claim.

**SCI-FLT-FIXED-REQ-041 - Single-convolution composition**

One SCI-FLT-FIXED v0.1 product MUST apply one exact resolved sampled convolution exactly once. Intermediate transformations, multiple application, selector collapse, or operator reordering MUST NOT be claimed by the product.

**SCI-FLT-FIXED-REQ-042 - Numerical-comparison policy**

Any future finite-precision conformance evidence MUST preregister every comparison-policy field in Section 13 before candidate results are observed. Comparison bounds MUST NOT change after a failure, and the policy MUST NOT authorize filter science or claim validation, adequacy, or performance.

**SCI-FLT-FIXED-REQ-043 - Exposure lineage**

Every complete candidate MUST carry FLT-EXPOSURE-LINEAGE with the exact parent exposure identity or typed absence. FLT MUST create no physical exposure, MUST keep influence distinct, and MUST NOT authorize a convolved exposure plane as filtered-signal physical exposure.

**SCI-FLT-FIXED-REQ-044 - External-resolution ownership**

"Externally resolved" MUST mean complete before application under one exact FLT-owned plan. It MUST NOT transfer FLT policy, transformation, selector, application, or publication ownership to an unnamed producer.

**16. Falsifiable predictions**

Each prediction is conditional on an exact admitted parent and fully resolved operator unless the prediction explicitly tests unavailability.

### SCI-FLT-FIXED-PRED-001 - Identity operator

For the exact identity operator, FLT-SIG equals the parent signal on  $S_{out}$ . An available compatible response and covariance are unchanged on that domain.  $K_{req} = \{0\}$  and  $S_{out}$  equals the exact admitted finite parent-row domain. The result is `realized_identity` only after successful publication, not disabled and not an unparented copy.

### SCI-FLT-FIXED-PRED-002 - Zero operator

For the exact zero operator, every FLT-SIG value on its declared inherited admitted parent-support row domain is zero; the local fixed-state derivative and parent-payload conditional covariance contribution are zero there. A missing complete source basis or parent response remains typed unavailable, and total systematic uncertainty remains separate. The product is `realized_zero` only after publication, not disabled, invalid, or evidence of precision.

### SCI-FLT-FIXED-PRED-003 - Input scaling

For any finite scalar  $a$ , applying the same frozen  $A_{Theta,J}$  to  $a \cdot m$  produces  $a \cdot y$ . Changing parent membership or re-resolving  $J_{full}$  is outside this conditional-linearity comparison. Failure on the fixed domain falsifies strict linearity.

### SCI-FLT-FIXED-PRED-004 - Constant input and DC gain

For constant admitted input  $m_q = c$ , every full-footprint convolution row is  $c$  times the exact signed kernel sum, equivalently  $c$  times the declared DC gain. Constant preservation occurs exactly when that gain is one; no other normalization implies it.

### SCI-FLT-FIXED-PRED-005 - Impulse response

An admitted single-pixel unit impulse produces the exact sampled kernel shifted according to the declared center, orientation, handedness, phase, and indexing, then restricted by  $S_{out}$ . Any implicit recentering, interpolation, reversal, or periodic copy falsifies the operator identity.

### SCI-FLT-FIXED-PRED-006 - Parent-response composition

Applying the transformation to an exact compatible fixed-state response, already realized response companion, or parent full-procedure difference produces exactly its separately identified  $A_{Theta,J}$  composition. A kernel-only surrogate, a second application, a re-resolved selector, or a response using different edge, phase, or normalization rules fails.

### SCI-FLT-FIXED-PRED-007 - Signed kernel

A signed kernel follows its exact signed coefficients, including cancellation. Its geometric, signed, absolute, and squared support summaries remain unequal when the coefficients make them unequal; none may substitute for another.

### SCI-FLT-FIXED-PRED-008 - Zero-sum kernel

For a complete constant input and exact zero signed-sum kernel, every admitted output row is zero. This nulling does not by itself make the row unavailable, uncertain, source-free, or statistically significant.

### SCI-FLT-FIXED-PRED-009 - Full-footprint boundary

For every output row whose exact  $K_{req}$  footprint is complete, admitted, finite, and predicate-passing, the row is in  $S_{out}$ . Removing, invalidating, or making non-finite any one required parent location removes every dependent output row from  $S_{out}$  with an exact cause.

### SCI-FLT-FIXED-PRED-010 - Deferred edge methods

Any result that uses extension, wrapping, truncation, local support renormalization, inpainting, reflection, clamp, mirror, padding-based admission, edge completion, or value replacement is rejected as a v0.1 product even when it is finite or preserves a constant.



**SCI-FLT-FIXED-PRED-011 - Missing and non-finite input**

No output row depending on a missing, unavailable, non-admitted, or non-finite required parent value belongs to  $S_{out}$ . Such a row is unavailable rather than zero and records the applicable cause.

**SCI-FLT-FIXED-PRED-012 - Complete covariance transform**

For an exact available compatible  $C_{parent}$ , the published complete covariance equals  $A_{Theta, J} C_{parent} transpose(A_{Theta, J})$  on the declared row ordering. A different domain, ordering, or omitted required term fails the complete- covariance claim.

**SCI-FLT-FIXED-PRED-013 - Off-diagonal covariance from diagonal input**

For an explicitly independent-diagonal parent model, overlapping nonzero output rows produce the exact cross term  $sum_j A_{ij} A_{lj} V_j$ . Publishing only marginals cannot pass a complete-relative-to-that-model covariance check when this term is nonzero, and that conditional completeness says nothing about unknown real parent cross terms.

**SCI-FLT-FIXED-PRED-014 - Unavailable parent companion**

For a base signal request, unavailable parent response or covariance yields an honest unavailable state record without blocking an otherwise complete signal. The same state fails a corresponding response-qualified or covariance-qualified request. A zero array, diagonal guess, weight, denominator, or kernel-only surrogate fails.

**SCI-FLT-FIXED-PRED-015 - WCS or grid mismatch**

Any mismatch in parent and output WCS, frame, topology, metric, shape, pixel indexing, or pixel-area convention makes the same-grid route unavailable. Approximate equality or successful numerical resampling does not pass.

**SCI-FLT-FIXED-PRED-016 - Observation and coadd identity**

A filtered observation, filtered MAP coadd, and filtered JINC observation have distinct parent and successor identities even if their arrays are numerically equal. No coadd of filtered observations is an SCI-FLT-FIXED v0.1 product.

**SCI-FLT-FIXED-PRED-017 - Exact NOI parity**

For each compatible admitted NOI member, applying the exact frozen signal operator produces  $A_{Theta, J} M_b$  on the identical row domain. A member missing one frozen required footprint is unavailable there; a different coefficient, parameter, grid, support, edge rule, generation, or selector makes the route unavailable.

**SCI-FLT-FIXED-PRED-018 - Per-member re-resolution**

If any NOI member selects or re-resolves a kernel, cutoff, support, threshold, edge state, or other parameter, that member is rejected from the fixed-state ensemble rather than mixed with it.

**SCI-FLT-FIXED-PRED-019 - Disabled, identity, zero, and failure**

A disabled route is not\_produced; identity and zero each pass through a complete publication candidate and publication decision before emitting their separately parented realized products; and a required failure emits no complete product and propagates its cause. Collapsing any pair fails the lifecycle contract.

**SCI-FLT-FIXED-PRED-020 - Upstream unavailable parent**

At the launch state recorded by the admitted boundaries, an attempted ordinary numerical MAP or ToITEC JINC transformation remains unavailable unless a separately authorized upstream successor supplies every named gate. The FLT contract alone cannot make that route numerical.

**SCI-FLT-FIXED-PRED-021 - Low-pass claim completeness**

Removing any required low-pass transfer fact from an otherwise complete fixed convolution makes only the low-pass qualification unavailable. Retaining a low-pass label with an incomplete transfer specification fails the contract.

**SCI-FLT-FIXED-PRED-022 - Exact-zero coefficient dependency**

For a kernel whose declared representation contains an exact-zero coefficient at one geometric offset, the offset remains in base-v0.1  $K_{req}$ . Making that parent location missing, unavailable, non-finite, or predicate-failing removes the dependent row even though its arithmetic coefficient is zero. A floating threshold that changes this result falsifies the method identity.

**SCI-FLT-FIXED-PRED-023 - Zero-operator row support**

The exact zero operator produces rows on its explicitly declared inherited admitted parent-support domain and nowhere else. An empty required footprint that admits storage rows outside that domain, or a zero array with no parent support identity, fails the zero-operator contract.

**SCI-FLT-FIXED-PRED-024 - Independent low-pass transfer check**

For a qualified low-pass subtype, an independent exact sampled-transfer evaluation on the declared frequency grid and metric reproduces the declared DC gain, passband bounds, transition behavior, stopband attenuation, phase, and anisotropy. PRED-021 metadata completeness without this numerical agreement is insufficient.